

System Impact Study Report

For

***PJM Upgrade Request
Queue Position AD2-018***

Roseland-Cedar Grove 230kV

March 2018

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Calpine Mid-Atlantic Development, LLC, the Interconnection Customer (IC), has proposed a merchant network upgrade of 63 MVA to the Roseland – Cedar Grove 230 kV line. The proposed in-service date for this project is June 1, 2019. **This study does not imply a PSE&G commitment to this in-service date.**

Cost Summary

The AD2-018 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrades	\$ 0
Non Direct Connection Network Upgrades	\$ 0
Allocation for New System Upgrades	\$ 0
Contribution for Previously Identified Upgrades	\$ 0
Merchant Network Upgrades	\$ 22,316,857
Total Costs	\$ 22,316,857

Merchant Network Upgrades

PJM NUN n5564	Estimate			Description of Reinforcement	New or Corrected Rating (MVA)	
	Cost	Tax	Time		Rate A	Rate B
Cost to increase Rate B by 63 MVA:	\$19,092,187	\$3,224,670	1 Yr	Reconductor 1590 ACSS	934	1080

ASSUMPTIONS

- This is a desktop study only and the final cost shall be determined by the facilities study.
- No site visits were performed.
- Desktop does not fully address the condition of the towers.
- No additional reinforcement scope beyond what is included in Conceptual Design.
- Duration of engineering, permitting, and construction is assumed to be 1 year.

Final ICTRs Granted for AD2-018:

Incremental Capacity Transfer Rights (ICTRs) were calculated using the 2021/22 RPM case. The following table provides the total ICTRs into the indicated Locational Deliverability Area (LDA) with the assumption that all active higher ordered queue projects were modeled in the 2021/22 RPM case. If there are any material changes associated with prior queued projects in the AD2 queue, then PJM would retool the analysis and determine the final ICTR's granted for the AD2-018 Merchant Transmission Request:

LDA/Ties	ICTRs
PSEG	179
PS North	69

The New Service Customer's rights as described above shall become effective upon the completion of the Network Upgrades, Local Upgrades, or Merchant Network Upgrades, and the rights described above are contingent upon the successful completion of upgrades identified for the higher ordered queue projects. The ICTRs may change if there are changes to the higher ordered queue projects such as project withdraws, or other project modifications as described in PJM OATT §205.5. PJM would determine if a re-study is required and inform the New Service Customer.